

Yufeng Gu

📍 4844 Bob and Betty Beyster Building, 2260 Hayward
Ann Arbor, MI, USA, 48109-2121
✉ yufenggu@umich.edu 📞 +1-734-272-5271

Education

University of Michigan

Ph.D., Computer Science and Engineering Department

Ann Arbor, MI, USA

Sep. 2020 - Apr. 2026

Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering

Hangzhou, China

Sep. 2016 - June 2020

Professional Experience

University of Michigan

Graduate Research Assistant, Advisor: Reetuparna Das

Ann Arbor, MI, USA

Sep. 2020 - Dec. 2025

Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

Tenstorrent Inc.

Performance Architect Intern, Manager: Wei-han Lien

Santa Clara, CA, USA

May 2023 - Aug. 2023

Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with <10% error rate. This model reduces the performance modeling time from a few hours to ~5 minutes, enabling fast kernel development and optimization.

Intel Labs

Graduate Research Intern, Manager: Nilesh Jain

Hillsboro, OR, USA

June 2022 - Aug. 2022

Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to $1.4\times$ request throughput improvement over single model deployment and reducing scheduling times from ~30 minutes to ~10 seconds compared to state-of-the-art schedulers.

Yale University

Undergraduate Research Intern, Advisor: James Duncan, Xiaoxiao Li

New Heaven, CT, USA

Nov. 2019 - Mar. 2020

Interpretation on ASD with fMRI and Deep Learning Models. Publications on MICCAI and MedIA.

École Polytechnique Fédérale De Lausanne (EPFL)

Undergraduate Research Intern, Advisor: Babak Falsafi

Lausanne, Switzerland

July 2019 - Sep. 2019

Patched QFlex Simulator (from Makefile to CMake compilation) and accelerated activation functions for DNN.

Award & Honors

Outstanding Young Speaker Award (APPT 2025)

Hardware-Software Co-Design for LLM Inference

July 2025

Distinguished Artifact Honorable Mention (HPCA 2025)

Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing. (3/29 Artifacts)

March 2025

Communication of ACM (CACM) Research Highlights

GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis.

CACM Research Highlights section selects 24/10,000+ papers from ACM conferences per year, reprinting the most significant and influential results across Computer Science.

July 2024

University of Michigan Rackham Graduate Student Research Grant (\$3000, role: PI)

Pangenomics Benchmark Suite and Characterization.

Apr. 2024

University of Michigan Rackham Travel Grant, ISCA/HPCA Travel Grant

2023, 2025

Fellowship of Summer@EPFL (2% applicants awarded)

July 2019

Tang Lixin Fellowship (2% students awarded)

Nov. 2017, 2018, 2019

Outstanding Student Leaders in Zhejiang University (3% students awarded)

Oct. 2017, 2019

First-Class Scholarship for Outstanding Students (2% students awarded)

Oct. 2017

Under-Review Publications

Yufeng Gu, Samel Gobriel, Nilesh Jain, Reetuparna Das. "DREAM: Data Reuse-Aware Processing-In-Memory Architecture for Efficient Large Language Model Inference." (*Under Submission*)

Yufeng Gu, Vui Seng Chua, Nilesh Jain, Ravishankar Iyer, Reetuparna Das. "Farm: Fast Resource Management for Quality of Service-Aware Co-Location of Machine Learning Inference." (*Under Submission*)

Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. "Addressing In-Memory Database Bottlenecks with CXL Memory Expansion." (*Under Submission*)

Peer-Reviewed Publications

* equal contribution

Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. "PangenomicsBench: A Benchmark Suite and Characterization of Pangenomics." To appear on IEEE International Symposium on Workload Characterization (IISWC) 2025.

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." (Invited Paper) Communications of the ACM, 2025.

Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 52th Annual International Symposium on Computer Architecture (ISCA'25).

Yufeng Gu*, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25).

Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*

Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

Talks

Architecture and System Co-Design for Scalable LLM Inference

PhD Forum at APPT 2025

July 2025

Los Alamos National Laboratory

May 2025

CXL-enabled PIM system for LLM inference

SAFARI Live Seminar (Host: Prof. Onur Mutlu)

June 2025

MCCSys workshop co-located with ASPLOS 2025

Mar. 2025

Genomics Benchmark Suite and Accelerator Design

Computer Architecture Seminar at UCF (Host: Prof. Di Wu)

Mar. 2024

Cornel University (Host: Prof. Christopher Batten)

Feb. 2024

Peisu Xia Forum at ICT, CAS

Dec. 2023

Teaching Experience

Programming at Discovery Engineering program at University of Michigan

July. 2022

Mathematics at Tuanlin Primary School, Guizhou, China

Aug. 2017

Services

Sub-reviewer for ISCA 2023, ISCA 2024, MICRO 2025.

Artifact evaluation reviewer for IISWC 2021, ISCA 2023, ISCA 2024, ISCA 2025.

Reviewer for University of Michigan graduate student admission

Jan. 2021, 2022

Mentorship

Dev Singhanian M.S. UM

Jan. 2025 - Apr. 2025

Zesen Zhao M.S. UM (Now Ph.D. at UMich)

Jan. 2025 - Aug. 2025

Wenjie Geng M.S. UM

Sep. 2024 - Aug. 2025

Mayne Mei B.Eng. UM (Now Engineer at Etched)

Jan. 2024 - Apr. 2024

Ao Luo M.S. UM

Jan. 2024 - June 2024

Yuzhe Ruan B.S. UM (Now M.S. at Yale)

Sep. 2023 - Apr. 2024

Donglin Yu B.Eng. (Now M.Eng. at UIUC)

Jan. 2023 - Apr. 2024

Dawit Melka B.Eng. Addis Ababa University (Now Engineer at iCog Labs)

May 2022 - Aug. 2022

James Gu B.Eng. UM (Now Engineer at Amazon)

May 2022 - Aug. 2022